

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

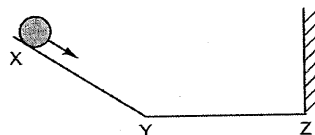
1. [ACJC/Prelims 2014/P3/4]

Two diamonds begin free fall from rest from the same height 1.0 s apart.
How long after the first diamond begin to fall will the two diamonds be 10 m apart?

- A 0.43 s B 1.43 s C 1.52 s D 2.43 s

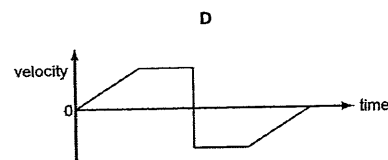
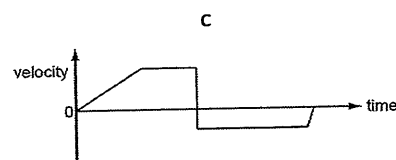
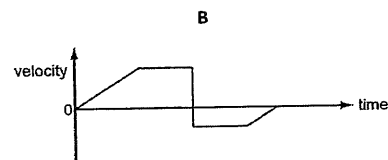
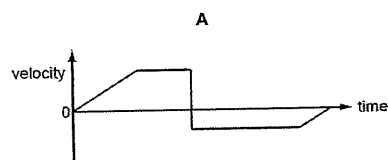
2. [ACJC/Prelims 2014/P1/4]

A ball is released from rest on a smooth slope XY.



It moves down the slope, along a smooth surface YZ and rebounds inelastically at Z. Then it moves back to Y and comes to rest momentarily somewhere on XY.

Which velocity-time graph represents the motion of the ball?



3. [AJC/Prelims 2014/P1/4]

A golf ball travels with an initial speed v_0 and it went only one-third of the way to the hole.

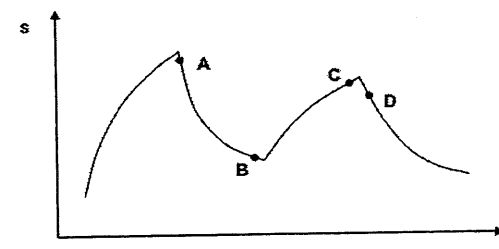
Assuming the force of resistance due to the grass remains the same, what new initial speed should the ball travel for the ball to go into the hole?

- A $\sqrt{2} v_0$ B $\sqrt{3} v_0$ C $\sqrt{6} v_0$ D $3v_0$

4. [AJC/Prelims 2014/P1/5]

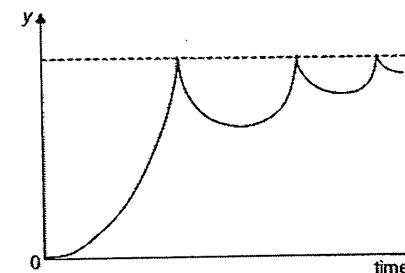
The graph is a displacement-time ($s-t$) graph for a tennis ball during part of a game.

Which part of the graph shows the highest speed?



5. [CJC/Prelims 2014/P1/4]

The graph below describes the motion of an object rebounding from a horizontal surface after being released from a point above the surface.



The quantity represented on the y-axis is the ball's

- A acceleration B displacement C momentum D velocity

6. [DHS/Prelims 2014/P1/4]

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a . Which quantities must be in the same direction?

- A a and v only B a and F only C v and F only D v , F and a

7. [HCI/Prelims 2014/P1/4]

Two stones, X and Y, of different mass are dropped from the top of a cliff. Stone Y is dropped a short time after stone X. Air resistance is negligible.

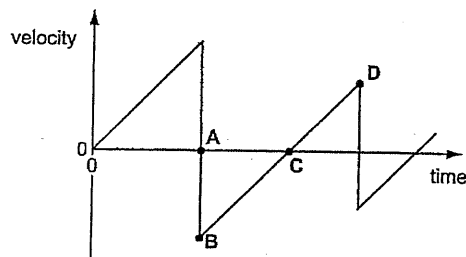
Whilst the stones are falling, the distance between them will

- A decrease if the mass of Y is more than the mass of X
- B increase if the mass of X is more than the mass of Y
- C decrease regardless if the mass of X more than or less than the mass of Y.
- D increase regardless if the mass of X is more than or less than the mass of Y

8. [JJC/Prelims 2014/P1/3]

A ball is released from rest above a hard, horizontal surface. The graph shows how the velocity of the bouncing ball varies with time.

At which point on the graph does the ball reach its maximum height after the first bounce?



9. [JJC/Prelims 2014/P1/4]

A lift is moving upwards with an acceleration of 4.0 m s^{-2} . A ball, held 2.5 m above the floor of the lift and at rest with respect to the lift, is released.

How long does it take for the ball to reach the floor of the lift?

- A 0.60 s
- B 0.71 s
- C 0.93 s
- D 1.1 s

10. [MJC/Prelims 2014/P1/4]

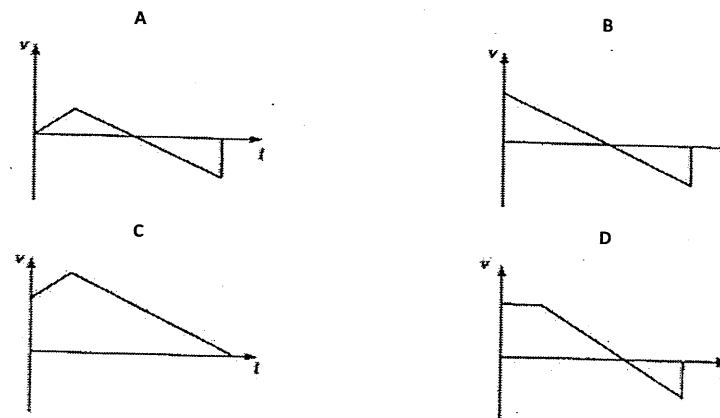
A stone is dropped from the top of a tower of height 50 m . The stone falls from rest and air resistance is negligible. What time is taken for the stone to fall the last 15 m to the ground?

- A 0.52 s
- B 1.7 s
- C 2.7 s
- D 3.2 s

11. [NJC/Prelims 2014/P1/4]

A toy rocket is launched vertically from Earth with a constant acceleration. After some time, the fuel is used up and the toy rocket falls freely back to Earth.

Which velocity-time graph best represents the journey? Neglect air resistance.



12. [PJC/Prelims 2014/P1/3]

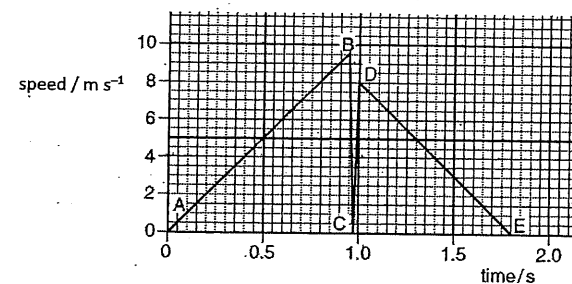
Car A travels horizontally at a uniform velocity of 15 m s^{-1} along a level road and passes car B, which is initially at rest, at time $t = 0 \text{ s}$. In order to catch up with car A, car B immediately accelerates uniformly along the same direction. At $t = 3 \text{ s}$, both cars are 27 m apart.

What additional time will it take for car B to catch up with car A?

- A 2.3 s
- B 4.5 s
- C 5.0 s
- D 7.5 s

13. [RVHS/Prelims 2014/P1/2]

A basketball falls vertically, hits the ground and rebounds vertically upwards. The diagram below is the speed-time graph for the ball.



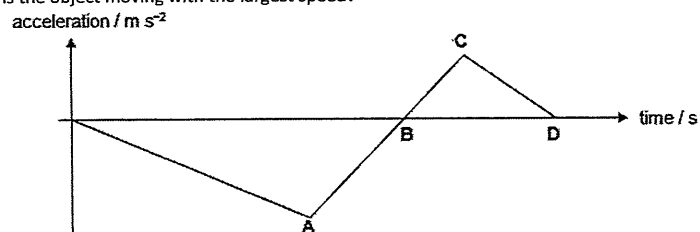
What is the overall change in velocity whilst the ball is in contact with the ground?

- A 0.49 m s^{-1}
- B 1.5 m s^{-1}
- C 18 m s^{-1}
- D 19 m s^{-1}

14. [SRJC/Prelims 2014/P1/3]

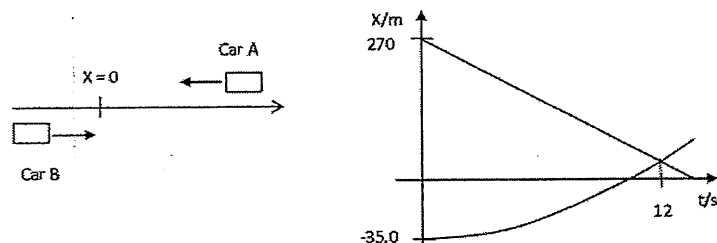
The acceleration-time graph of an object moving in a straight line is as shown. The object started its motion from rest.

At which point is the object moving with the largest speed?



15. [TPJC/Prelims 2014/P1/3]

A graph of displacement X against time t shows a car A and a car B moving towards each other.



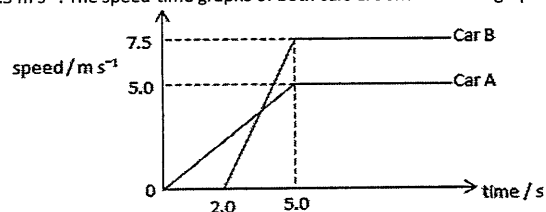
Car A has a constant speed of 20 m s^{-1} . Car B begins from rest at $t = 0$ with constant acceleration.

What is the acceleration magnitude of car B?

- A 0.511 m s^{-2} B 0.852 m s^{-2} C 0.903 m s^{-2} D 1.43 m s^{-2}

16. [VJC/Prelims 2014/P1/4]

Car A and car B both start from the same starting line. Car A moves off first, accelerating constantly for 5.0 s to a constant speed of 5.0 m s^{-1} . Car B starts moving 2.0 s after car A, and accelerates constantly for 3.0 s until it reaches a constant speed of 7.5 m s^{-1} . The speed-time graphs of both cars are shown in the graph below:



At what time does car B catch up with car A?

- A 3.3 s B 5.0 s C 5.5 s D 7.5 s

17. [VJC/Prelims 2014/P1/5]

A man is standing inside a descending lift.

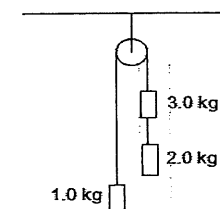
Which of the following statements about the magnitude of the force exerted on the man's feet by the floor of the lift is always correct?

- A It is less than the magnitude of his weight.
B It is equal to the magnitude of his weight.
C It would not be equal to what it would be in a stationary lift.
D It is equal to the magnitude of the force exerted on the floor of the lift by the man's feet.

18. [CJC/Prelims 2014/P1/8]

A light inextensible string is wound, as shown, over a frictionless, light pulley.

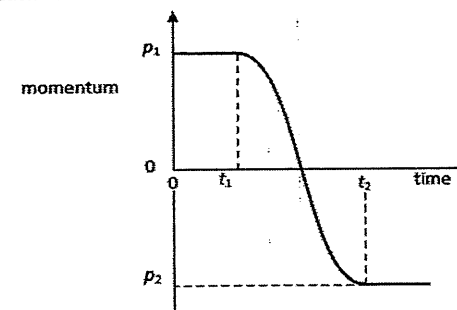
What is the tension in the string between the 2.0 kg and the 3.0 kg masses when the system is released?



- A 6.5 N B 16 N C 20 N D 39 N

19. [DHS/Prelims 2014/P1/8]

The graph shows the variation with time of the momentum of a ball as it is kicked in a straight line.



Initially, the momentum is p_1 at time t_1 . At time t_2 the momentum is p_2 .

What is the magnitude of the average force acting on the ball between times t_1 and t_2 ?

- A $\frac{p_1 - p_2}{t_2}$ B $\frac{p_1 - p_2}{t_2 - t_1}$ C $\frac{p_1 + p_2}{t_2}$ D $\frac{p_1 + p_2}{t_2 - t_1}$

20. [HCI/Prelims 2014/P1/6]

A man of mass M is standing on a weighing scale in an elevator. The elevator undergoes several different motions as described below.

Elevator's motion	Weighing scale reading
Moving downwards and coming to rest with uniform deceleration	N_1
Moving downwards and speeding up with constant acceleration	N_2
Moving downwards with constant speed	N_3

Which of the following is correct?

- A $N_1 = N_2 = N_3$ B $N_1 = N_2 < N_3$ C $N_1 < N_3 < N_2$ D $N_2 < N_3 < N_1$

21. [IJC/Prelims 2014/P1/5]

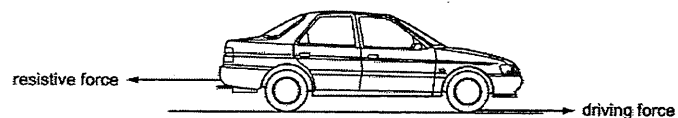
A tennis ball approaches a tennis player who gives it a hard hit with a swing of his racket, returning the ball to his opponent.

During the time of contact between the racket and the ball,

- A the force on the racket due to the ball is smaller than the force on the ball due to the racket because the ball is much smaller and lighter than the racket.
 B the force on the racket due to the ball is smaller than the force on the ball due to the racket because the ball moves off with high speed after the contact whereas the racket does not.
 C the force on the racket due to the ball is larger than the force on the ball due to the racket because smaller mass means smaller force.
 D the force on the racket due to the ball is equal to the force on the ball due to the racket.

22. [IJC/Prelims 2014/P1/6]

A car of mass 750 kg has a horizontal driving force of 2.0 kN acting on it. It has a forward horizontal acceleration of 2.0 m s^{-2} .



What is the resistive force acting horizontally?

- A 0.50 kN B 1.5 kN C 2.0 kN D 3.5 kN

23. [MJC/Prelims 2014/P1/5]

The diagram below shows 4 identical wooden blocks connected by inelastic strings, A, B and C. A constant force accelerates the blocks to the right on a horizontal frictionless table.

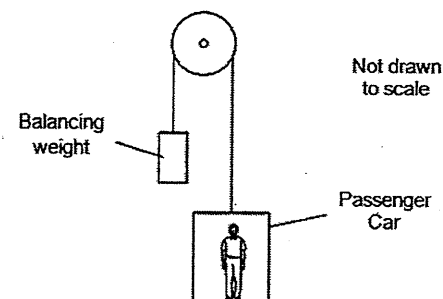


Which string has the greatest tension?

- A String A
 B String B
 C String C
 D All have the same tension

24. [MJC/Prelims 2014/P1/6]

An elevator consists of a passenger car of mass 520 kg, supported by a cable that runs over a light, frictionless pulley to a balancing weight of mass 640 kg. The balancing weight falls as the passenger car rises. The passenger has a mass of 80 kg.

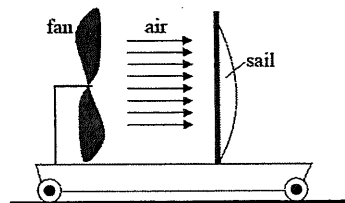


What is the magnitude of the acceleration of the passenger car as it rises?

- A 0.032 m s^{-2} B 0.32 m s^{-2} C 0.65 m s^{-2} D 0.75 m s^{-2}

25. [NJC/Prelims 2014/P1/7]

A fan and a sail are mounted vertically on a cart that is initially at rest on a horizontal table as shown in the diagram.



When the fan is turned on an air stream is blown towards the right and is incident on the sail. The cart is free to move with negligible resistant forces. After the fan has been turned on the cart will

- A move to the left and then to the right.
- B remain at rest.
- C move towards the right.
- D move towards the left.

26. [NYJC/Prelims 2014/P1/5]

A box of weight W is being pushed, by a force, F up a frictionless surface inclined at an angle α . F is directed at an angle of β below the horizontal.

What is the magnitude of the acceleration of the box?

- A $\frac{[F \cos(\alpha + \beta) - W \sin \alpha]g}{W}$
- B $\frac{[F \cos(\alpha - \beta) - W \sin \alpha]g}{W}$
- C $\frac{[F \cos \alpha \cos \beta - W \sin \alpha]g}{W}$
- D $\frac{[F \sin \alpha \sin \beta - W \sin \alpha]g}{W}$

27. [NYJC/Prelims 2014/P1/6]

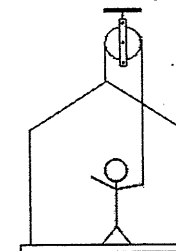
An 80.0 kg mass falls from a 3.00 m tall ledge down onto the ground. It hits a hard surface and comes to rest in 0.0250 s.

What is the average force exerted by the surface on it?

- A 19.6 N
- B 785 N
- C 24 600 N
- D 25 300 N

28. [PJC/Prelims 2014/P1/6]

A man raises himself and the crate on which he stands with an acceleration of 2.0 m s^{-2} with a massless rope and frictionless pulley system as shown. The mass of the man is 80 kg and that of the crate is 50 kg.

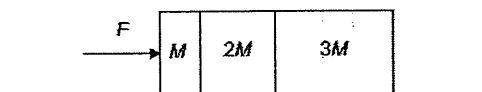


What is the tension in the rope?

- A 590 N
- B 770 N
- C 940 N
- D 1450 N

29. [RVHS/Prelims 2014/P1/5]

Three blocks with masses M , $2M$ and $3M$ are pushed along a rough surface by a horizontal force F as shown.



The friction between mass M and the rough surface is f , and f is proportional to M .

What is the magnitude of the force that mass $3M$ exerts on mass $2M$?

- A $\frac{F}{2}$
- B $\frac{F}{2} + 3f$
- C $\frac{F}{3} + f$
- D $\frac{F}{3} + 3f$

30. [SAJC/Prelims 2014/P1/7]

An object of mass m is hanging by a string from the roof of a lift. The lift is moving upwards but is slowing down. The tension in the string is

- A less than mg .
- B exactly mg .
- C greater than mg .
- D zero.

31. [SRJC/Prelims 2014/P1/6]

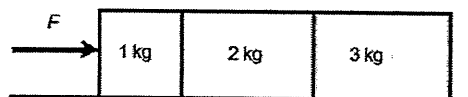
A truck of mass 2300 kg has broken down on a road. A car of mass 1200 kg decides to give it a push from the back and accelerates at a rate of 3.0 m s^{-2} . The average frictional force that the car and truck experience is 350 N and 800 N respectively.

What is the driving force of the car?

- A 3750 N
- B 3950 N
- C 11 300 N
- D 11 650 N

32. [TJC/Prelims 2014/P1/8]

Three boxes of masses 1 kg, 2 kg, and 3 kg are pushed along a smooth surface by a force F .

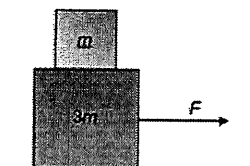


What is the force that the 2 kg mass exerts on the 1 kg mass?

- A $\frac{F}{3}$ B $\frac{F}{2}$ C $\frac{3F}{4}$ D $\frac{5F}{6}$

33. [TPJC/Prelims 2014/P1/6]

A box of mass m rests on another box of mass $3m$. The more massive box slides along a frictionless surface due to a pulling force F . What is the friction between the boxes?



- A F B $\frac{F}{2}$ C $\frac{F}{3}$ D $\frac{F}{4}$

34. [VJC/Prelims 2014/P1/5]

A man is standing inside a descending lift.

Which of the following statements about the magnitude of the force exerted on the man's feet by the floor of the lift is always correct?

- A It is less than the magnitude of his weight.
 B It is equal to the magnitude of his weight.
 C It would not be equal to what it would be in a stationary lift.
 D It is equal to the magnitude of the force exerted on the floor of the lift by the man's feet.

35. [YJC/Prelims 2014/P1/5]

A man is parachuting at constant speed towards the surface of the Earth. The force which, according to Newton's third law, makes an action-reaction pair with the weight on the man is

- A the drag force acting on the man due to air resistance.
 B the gravitational force on the Earth due to the man.
 C the gravitational force on the man due to Earth.
 D the tension in the harness of the parachute.

36. [YJC/Prelims 2014/P1/6]

A trolley runs down a slope with a constant acceleration a . The mass of the trolley is now tripled and the trolley is allowed to run down the same slope. In both cases, effects of friction and air resistance are negligible.

Which statement is correct for the second experiment?

- A The accelerating force is the same.
 B The acceleration is $\frac{1}{3}a$.
 C The acceleration is a .
 D The acceleration is $3a$.

II. Structured Questions

1. [MJC/Prelims 2014/P3/1]

A small wooden block is projected up a frictionless ramp with an initial speed of 4.0 m s^{-1} as shown in Fig. 1.1 below. The ramp is inclined at 20° and the block was projected from a point 6.0 m from the bottom of the ramp.

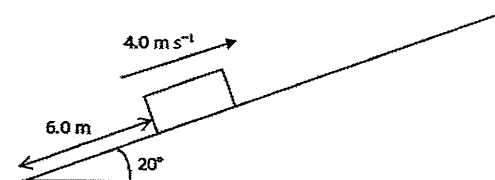


Fig. 6.1

- a. Calculate the distance moved by the block before it comes to instantaneous rest. [2]
 b. Determine the time taken for the block to reach the bottom of the ramp from the time it was projected. [2]
 c. Sketch the displacement-time graph of the entire motion from the time the block was projected to the time it reaches the bottom of the ramp. [2]

[Total: 6]

2. [RVHS/Prelims 2014/P3/1]

- a. Estimate the mass of an empty soft drink can. [1]
 b. Estimate the height of a window on the 20th floor of a HDB block of flats from the ground. [2]
 c. A person staying on the 20th floor drops an empty soft drink can out from his window. Estimate, with reason, the maximum speed with which the can strikes the ground. State 2 simplifying assumptions you made in your calculations. [4]

[Total: 7]

3. [TJC/Prelims 2014/P2/1]

A hot air balloon was rising steadily at a speed of 10.0 m s^{-1} .

A sandbag was dropped from the balloon at time $t = 0 \text{ s}$.

- a. Ignoring air resistance and taking g as 10 m s^{-2} , sketch on Fig. 3.1 the variation of the velocity v with time t of the sandbag from time $t = 0 \text{ s}$ to $t = 5.0 \text{ s}$. Label this graph A. Appropriate values should be indicated. [2]

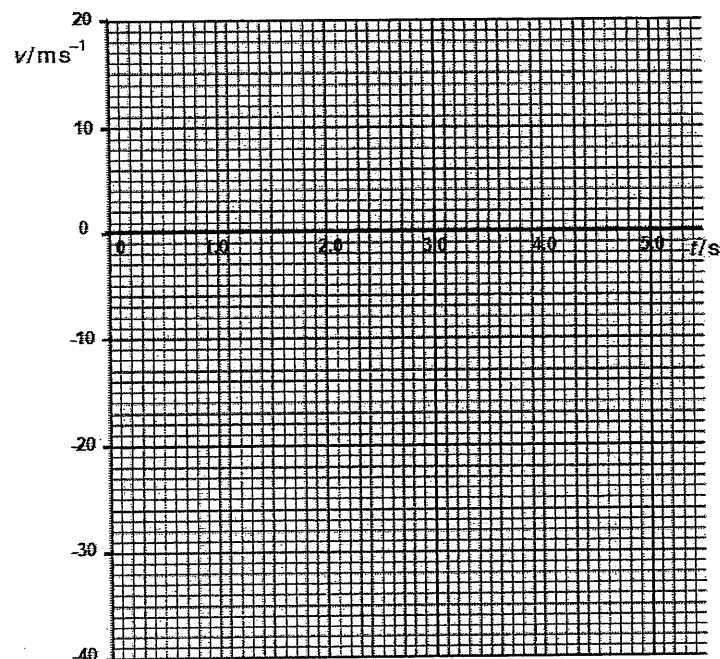


Fig. 3.1

- b. Another sandbag was dropped from the balloon at $t = 2.0 \text{ s}$. On the same axes provided in Fig. 3.1, sketch the variation of the velocity v with time t of the second sandbag from time $t = 0 \text{ s}$ to $t = 5.0 \text{ s}$. Label this graph B. Appropriate values should be indicated. [2]
- c. Using the graphs in Fig. 3.1 or otherwise, calculate the vertical distance between the two sandbags at $t = 5.0 \text{ s}$. [3]

[Total: 7]

4. [AJC/Prelims 2014/P2/2]

Fig. 4.1 shows a signboard suspended by two elastic ropes of tension T_1 and T_2 . The tension in T_1 is 300 N and the tension in T_2 is 252 N .

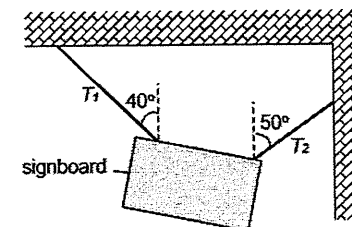


Fig. 4.1

- a. i. State the conditions for equilibrium. [2]
- ii. On Fig. 4.1, mark the centre of gravity of the signboard with a dot and label the point as G. Show clearly your construction to determine the centre of gravity on Fig. 4.1. [1]
- iii. Determine the weight of the signboard. [1]
- b. The signboard is pulled vertically downwards with a force of 20 N so that the ropes are stretched. Determine the acceleration of the signboard immediately after it is released. [2]

[Total: 6]

5. [TPJC/Prelims 2014/P2/1]

Two blocks, P and Q, of masses 0.30 kg and 1.50 kg respectively, are connected by a string that passes over a pulley as shown in Fig. 5.1. The pulley is frictionless and the string is inextensible.

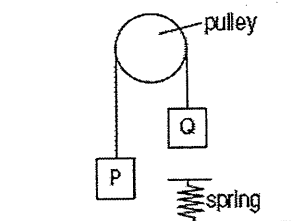


Fig. 5.1

The system is released from rest. Block Q falls vertically before it strikes a spring that is firmly attached to the floor. The spring constant is 500 N m^{-1} .

- a. i. Using the space provided below, draw the free-body diagrams of Blocks P and Q at the instant when the system is released from rest. [2]
- ii. Show that the magnitude of acceleration of Block Q is 6.54 m s^{-2} before striking the spring. [2]
- iii. Hence, determine the tension in the string before Block Q strikes the spring. [1]
- b. The acceleration of Block Q decreases after it touches the spring. Eventually Block Q comes to a stop and stays at rest on the spring. Determine the compression of the spring. [3]

[Total: 8]

	Raffles Institution Physics Department Revision Package 2	Motion and Forces	3
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III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	A	B	A	B	B	D	C	A	A	A	B	C	B	C

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	D	A	B	D	D	A	C	B	B	A	D	B	A	A

31	32	33	34	35	36
D	D	D	D	B	C

1 C Solving $\frac{1}{2}gt^2 - \frac{1}{2}g(t-1)^2 = 10$ gives $t = 1.52$ s

2 A The 2 slopes have the same gradient since same magnitude of acceleration.
The time taken to travel back to slope is longer after collision since speed after rebound is lower.

3 B $0 = v_o^2 - 2a\left(\frac{1}{3}s\right) \Rightarrow v_o^2 = 2a\left(\frac{1}{3}s\right)$ (1) OR using $\frac{1}{2}mv^2 = Fx$
 $0 = u^2 - 2as \Rightarrow u^2 = 2as$ (2)
 $(1)/(2): \frac{v_o^2}{u^2} = \frac{1}{3}$
 $u = \sqrt{3} v_o$

4 A A has the steepest gradient.

5 B The point of release is taken to be $y = 0$ and the horizontal surface from which the ball rebounds from is represented by the dotted line on the graph. The displacement of the ball is observed to decrease with time for subsequent bounces.

Raffles Institution

Revision Package (H2 Physics)

3 – Motion and Forces

- 6 B The acceleration of the object is due to the resultant force acting on it. Hence both of them should always be in the same direction.
The rest of the options are wrong because force / acceleration and velocity need not always be in the same direction. When force is opposite in direction to velocity, it means that the object is slowing down.
- 7 D Since air resistance is negligible, the acceleration of the stones is the acceleration due to gravity, which is independent on mass. Since stone X is dropped first, stone X will always have a higher speed than stone Y. Therefore X will always move a larger distance at every second compared to Y.
- 8 C At the maximum height, it should be a turning point. As such, the velocity at that moment should be zero. Between A and C, A is the point at which the ball hits the floor momentarily to switch directions (which explains why it is at the intersection of positive and negative region) and the vertical line suggests a sudden change. Prior to that change, the ball is moving downwards (dropped) represented by a positive value. C is the point at which the ball goes from negative velocity to positive velocity (with values) which suggests moving upwards to downwards.
- 9 A Taking upwards as positive,
Acceleration of ball relative to Earth, $a = 4.0 - (-9.81) = 13.81 \text{ m s}^{-2}$
 $s = ut + \frac{1}{2}at^2$
 $2.5 = 0 + \frac{1}{2}(13.81)t^2$
 $t = 0.60 \text{ s}$
- 10 A Using $s = ut + \frac{1}{2}at^2$
 $50 = 0 + \frac{1}{2} \times 9.81 \times t_1^2 \Rightarrow t_1 = 3.193 \text{ s}$
 $50 - 15 = 0 + \frac{1}{2} \times 9.81 \times t_2^2 \Rightarrow t_2 = 2.671 \text{ s}$
 $t_2 - t_1 = 3.193 - 2.671 = 0.522 \text{ s}$
- 11 A Taking direction upwards as positive, the velocity of the rocket increases at a constant in the positive direction initially. After the fuel is used up, the rocket will still move upwards but slows down at a constant rate of 9.81 m s^{-2} until it reaches zero velocity, which is the highest point of its motion. It will then fall back towards Earth at a constant rate of 9.81 m s^{-2} . Since the rocket is moving downwards, its velocity will be negative during its downward motion.
- 12 B At $t = 3 \text{ s}$, $s_A - s_B = 27 \Rightarrow (15 \times 3) - \left(\frac{1}{2} \times a_B \times 3^2\right) = 27 \Rightarrow a_B = 4.0 \text{ ms}^{-2}$
Let the time for car B to catch up with car A be t' .
 $15 \times t' = \frac{1}{2} \times 4.0 \times t'^2 \Rightarrow t' = 7.5 \text{ s}$
Therefore, the additional time required = $7.5 - 3.0 = 4.5 \text{ s}$

13 C Change in velocity = $8 - (-9.5) = 17.5 \text{ m s}^{-1}$

14 B Area under $a-t$ graph represents the change in velocity. Hence, the speed at point B will be greatest.

15 C $s_A + s_B = 270 - (-35) \Rightarrow (20)(12) + \frac{1}{2}a(12)^2 = 305 \Rightarrow a = 0.903 \text{ m s}^{-2}$

16 C Car B catches up with car A when both travel the same distance. So find the time when this happens.
At $t = 5.0 \text{ s}$:

Distance travelled by Car A = area under car A's graph = $0.5 \times 5.0 \times 5.0 = 12.5 \text{ m}$

Distance travelled by car B = $0.5 \times 3.0 \times 7.5 = 11.25 \text{ m}$

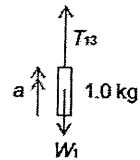
So after $t = 5.0 \text{ s}$, car B must travel $12.5 - 11.25 = 1.25 \text{ m}$ more than car A.

Time needed to do this = $1.25 / (7.5 - 5.0) = 0.5 \text{ s}$

So car B catches up with car A at $t = 5.0 + 0.5 = 5.5 \text{ s}$

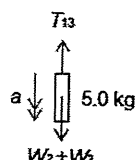
17 D By Newton's Third Law, the action and reaction pair is equal in magnitude.

18 A



$$T_{13} - W_1 = m_1 a$$

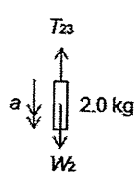
$$T_{13} = a + g$$



$$W_2 + W_3 - T_{13} = (m_2 + m_3) a$$

$$5g - T_{13} = 5a$$

$$5g - (a + g) = 5a \Rightarrow a = \frac{2}{3}g$$



$$W_2 - T_{23} = m_2 a$$

$$T_{23} = 2g - 2a = 2g - 2\left(\frac{2}{3}g\right)$$

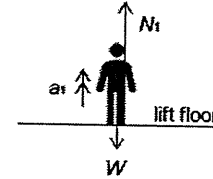
$$= 6.54 \text{ N}$$

19 B The force only acts during t_1 to t_2 (as the change in momentum only occurs then).

The total change in momentum = $(p_2 - p_1)$ where p_2 is a negative number.

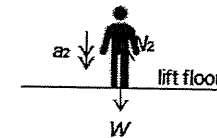
ave $F = \frac{\Delta p}{\Delta t} = \frac{p_2 - p_1}{t_2 - t_1}$. This will result in a negative answer.

In order to obtain the magnitude, $|\text{ave } F| = \left| \frac{\Delta p}{\Delta t} \right| = \left| \frac{p_2 - p_1}{t_2 - t_1} \right| = \frac{p_1 - p_2}{t_2 - t_1}$



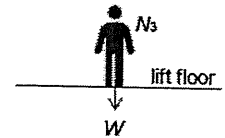
$$N_1 - W = ma_1$$

$$N_1 = W + ma_1$$



$$W - N_2 = ma_2$$

$$N_2 = W - ma_2$$



$$N_3 = W$$

Since weight of the man is constant, $N_1 > N_3 > N_2$

21 D The force on the racket must be equal and opposite to the force on the ball based on Newton's 3rd Law.

22 A By Newton's 2nd Law,

$$F_{\text{driving}} - F_{\text{resistive}} = ma$$

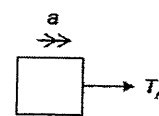
$$F_{\text{resistive}} = 2000 - (750)(2.0) = 500 \text{ N}$$

23 C For all the masses as a whole, $F = m_T a = 4ma$

Since all the masses move together, they have the same acceleration a .

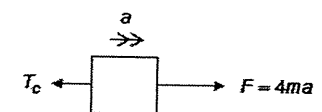
The greatest tension is likely to happen at the extreme ends.

Left most block



$$F_{\text{net}} = ma \Rightarrow T_A = ma$$

Right most block



$$F_{\text{net}} = F - T_C = ma \Rightarrow T_C = 3ma$$

24 B For the balancing weight

$$W_B - T = m_B a$$

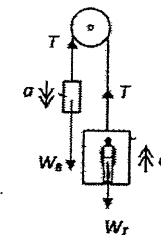
$$640g - T = 640a$$

For the elevator & passenger,

$$T - W_A = m_A a$$

$$T - 600g = 600a$$

Solving for simultaneous equations: $a = 0.32 \text{ m s}^{-2}$



25 B When the fan is switched on, it pushes the air. The air in turn exerts a force on the sail. By Newton's 3rd Law, the sail exerts an equal and opposite force on the air, which pushes back onto the fan.

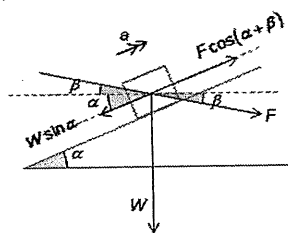
Since they are part of the same system, both forces act on the same system and cancel out. The cart thus remains at rest.

26 A

$$F_{\text{net}} = ma$$

$$F \cos(\alpha + \beta) - W \sin \alpha = \left(\frac{W}{g}\right)a$$

$$a = \frac{(F \cos(\alpha + \beta) - W \sin \alpha)g}{W}$$



27 D $v^2 = u^2 + 2as = 0 + 2(9.81)(3.00) \Rightarrow v = 7.672 \text{ m s}^{-1}$

By Newton's 2nd Law

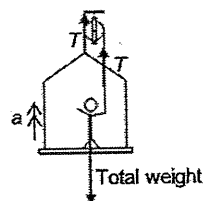
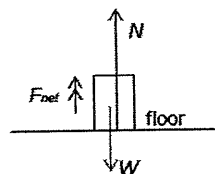
$$F_{\text{net}} = \frac{\Delta p}{\Delta t} = \frac{80(0 - 7.672)}{0.0250} = -24550 \text{ N}$$

This implies that the net force on the object is upwards.

$$N - W = F_{\text{net}}$$

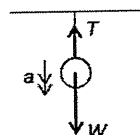
$$N - 80 \times 9.81 = 24550$$

$$N = 25300 \text{ N}$$



29 A Consider the whole system, $F - 6f = 6Ma$
Considering just the $3M$ block, $N - 3f = 3Ma$
Solving for simultaneous equations, $N = \frac{F}{2}$

30 A $W - T = ma$
 $T = mg - ma$



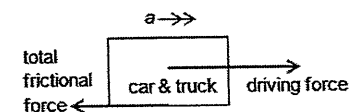
31 D Considering the car and truck as one system

$$F_d - f = Ma$$

$$F_d = Ma + f$$

$$= (2300 + 1200)(3.0) + 350 + 800$$

$$= 11650 \text{ N}$$

32 D Consider the whole system, $F = 6a$

Considering the 2 kg & 3 kg blocks as one system, $N = 5a \Rightarrow N = \frac{5}{6}F$

33 D Considering the whole system, $F = 4ma$

Considering just the top box, friction is the only force acting on it horizontally. Hence if it were to move together with the bottom box, it is due to the friction between the two boxes.

$$f = ma \Rightarrow f = \frac{3}{4}F$$

34 D By Newton's 3rd Law, the force exerted on the man by the floor will be equal and opposite to the force exerted on the floor by the man.

Options A to C are contingent on which direction the lift is accelerating in.

35 B Action-reaction pairs have to be of the same type and act on different objects.

Since the question is asking for weight which is the gravitational force on the man by the Earth, the reaction pair to this force has to be a gravitational force exerted by the man on the Earth.

36 C Accelerating force or net force is dependent on mass ($F = ma$).

Whereas the acceleration along the slope is a component of the acceleration of free fall (which is a constant).

Structured Questions

1. [MJC/Prelims 2014/P3/1]

a. Taking up the slope as positive,

$$a = g \sin \theta = -3.355$$

$$v^2 = u^2 + 2as$$

$$0 = 4^2 + 2(-3.355)(s)$$

$$s = 2.38 \text{ m}$$

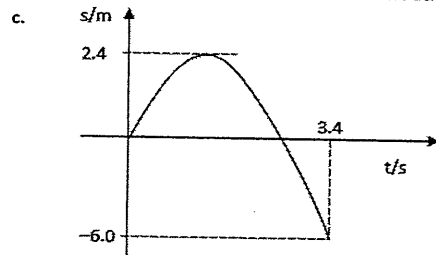
b.

$$s = ut + \frac{1}{2}at^2$$

$$-6 = 4t + \frac{1}{2}(-3.355)t^2$$

$$1.6775t^2 - 4t - 6 = 0$$

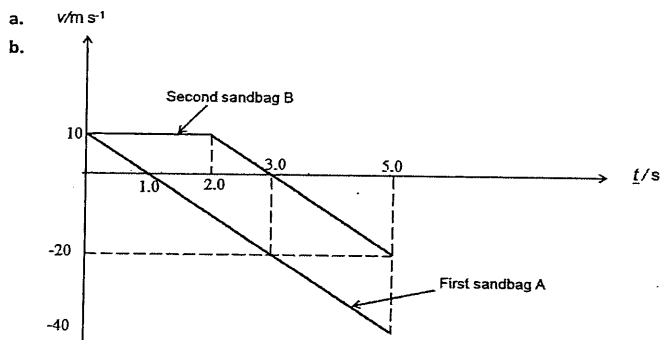
$$t = 3.43 \text{ s or } -1.04 \text{ s (inadmissible)}$$



2. [RVHS/Prelims 2014/P3/1]

- a. 20 to 80 g
- b. height of one storey = 2.5 to 3.1 m (typical floor to floor height for HDB = 2.8 m)
height $\times 20$ or height $\times 19 + (\text{height}/2)$
49 to 62 m
- c. Assume no air resistance/upthrust.
Assume initial speed of can is zero.
Assume $g = 9.81 \text{ m s}^{-2}$ is constant.
Not accepted:
- Drink can does not reach terminal velocity. Reason: if air resistance is negligible, can would not reach terminal velocity.
 - Drink can falls vertically downwards. Reason: if initial speed is zero, then it will fall vertically downwards.
 - Only force acting on drink can is gravitational force. Reason: the main point should be acceleration has to be constant in order to use the kinematics equation, whereas even if the only force is weight, g can vary with distance from ground.
 - No wind, no obstacles. Reason: question asked for simplifying assumptions made in calculations. Not very relevant to question.
- $u = 0, g = 9.81, s = \text{height}, v^2 = u^2 + 2as = 2 \times 9.81 \times \text{height} \Rightarrow v = 30 \text{ to } 35 \text{ m s}^{-1}$
- Alternative method: Decrease in GPE = Increase in KE

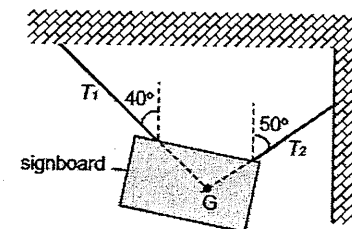
3. [TJC/Prelims 2014/P2/1]



- c. For first sandbag after $t = 5.0 \text{ s}$, displacement, $s_1 = 10 \times 5.0 - \frac{1}{2}(10)(5.0^2) = -75 \text{ m}$
For 2nd sandbag, displacement, $s_2 = \text{area under graph} = 10 \times 2.0 + \frac{1}{2}(10)(1.0) - \frac{1}{2}(20)(2.0) = +5.0 \text{ m}$
(Alternative: $s_2 = 10 \times 2.0 + [10 \times 3.0 - \frac{1}{2}(10)(3.0^2)] = 5.0 \text{ m}$)
Hence, total distance between them = $75 + 5.0 = 80 \text{ m}$

4. [AJC/Prelims 2014/P2/2]

- a. i. See definitions in lecture notes.
- ii. G is at the intersection of lines of action of T_1 and T_2



- iii. Vertically,

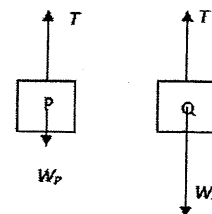
$$W = T_1 \cos 40^\circ + T_2 \cos 50^\circ = 300 \cos 40^\circ + 252 \cos 50^\circ = 391.8 = 390 \text{ N}$$

- b. Immediately after release, $F_{\text{net}} = ma = 20 \text{ N}$

$$\therefore a = \frac{F_{\text{net}}}{m} = \frac{F_{\text{net}}g}{W} = \frac{20(9.81)}{(391.8)} = 0.501 \text{ m s}^{-1}$$

5. [TPJC/Prelims 2014/P2/1]

- a. i.



Note:
 $W_P < T$
 $T < W_Q$
 T are of equal length

- ii. Consider P as system, $T - W_P = m_P a \quad \text{--- (1)}$
Consider Q as system, $W_Q - T = m_Q a \quad \text{--- (2)}$
Equation (1) + (2), $(1.50)(9.81) - (0.30)(9.81) = (1.50 + 0.30)a$
 $a = 6.54 \text{ m s}^{-2}$

- iii. Equation (1): $T - 0.30(9.81) = 0.30(6.54)$
 $T = 4.91 \text{ N}$

- b. After Block Q comes to a stop, the forces acting on P and Q are in equilibrium.
Consider P as system, $T = W_P \quad \text{--- (1)}$
Consider Q as system, $T = W_Q - kx \quad \text{--- (2)}$
Solving, $x = 0.0235 \text{ m}$